

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 31 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties of Definite Integrals & King's Rule	31 PYQs	Definite Integral Properties

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties of Definite Integrals & King's Rule	$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$, $\int_0^{\pi/2} \frac{\sin^n x}{\cos^n x} dx = \frac{\pi}{4}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties of Definite Integrals & King's Rule (Very High Yield — 31 PYQs)

Core Principle: King's property: $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$. For symmetric limits: $\int_{-a}^a f(x)dx = 0$ if $f(x)$ is odd, and $2\int_0^a f(x)dx$ if $f(x)$ is even. Integral of periodic functions: $\int_0^{nT} f(x)dx = n\int_0^T f(x)dx$.

■ **Caution / JEE Pitfall:** Applying the odd function property $\int_{-a}^a f(x)dx = 0$ without verifying that the limits are exactly symmetric $[-a, a]$.

■ **Speed Trick:** Any integral of the form $\int_a^b (f(x) / f(x) + f(a+b-x)) dx$ is directly equal to $(b-a) / 2$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int_{-6}^6 (a-6)^b - 6f(x+6)dx$ is equal to..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "One of the possible functions $f(x)$ which satisfies $\int_{-2}^2 f(x)dx=0$ is..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): " $\int_0^1 \log((1/x)-1)dx$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$\int_{a-6}^{b-6} f(x+6) dx$ is equal to

- (A) $\int_a^b f(x+6) dx$
- (B) $\int_a^b f(x-6) dx$
- (C) $\int_a^b f(x) dx$ **[Correct]**
- (D) $-\int_a^b f(x) dx$

Correct Answer: C

Solution Analysis: Put $t=x+6$ Then $dt=dx$ Now change the limits. When $x=a-6$, we get $t=(a-6)+6=a$ When $x=b-6$, we get $t=(b-6)+6=b$ So, $\int_{a-6}^{b-6} f(x+6) dx = \int_a^b f(t) dt$ The letter used for the variable does not matter in a definite integral. So, $\int_a^b f(t) dt = \int_a^b f(x) dx$ Hence, $\int_{a-6}^{b-6} f(x+6) dx = \int_a^b f(x) dx$ So the correct option is Option C